

Samia Jaman

929-530-6933 | sjaman@vassar.edu | [linkedin.com/in/sami-jaman](https://www.linkedin.com/in/sami-jaman) | github.com/samijaman

EDUCATION

Vassar College

Bachelor of Arts in Computer Science; Correlates in Mathematics and Art History

Poughkeepsie, NY

Aug. 2022 – Dec. 2026

- **Relevant Coursework:** Natural Language Processing, Probability, Statistical Inference, Linear Algebra, Data Structures & Algorithms, Theory of Computation, Software Implementation and Design, Operating Systems, Object-Oriented Programming
- **Affiliations:** ColorStack, Varsity Rugby, AMR Agency Talent

RESEARCH

Differentiating AI-Generated Text from Human-Written Text Across LLMs

Fall 2025

Python, PyTorch, DistilBERT, scikit-learn, Pandas

- Fine-tuned a DistilBERT classifier with a class-weighted loss over a 1,460-sample corpus containing only 6% AI-generated text, reaching **97.3% overall accuracy** (99.6% on human-written, 95.0% on AI-generated) and correctly labeling 100% of held-out essays from four unseen models (Claude, Gemini, Perplexity, GPT-5) at 0.9937–0.9969 confidence.
- Benchmarked against Logistic Regression baselines using Bag-of-Words and TF-IDF vectorizers: both scored F1 of 1.00 and 0.99 in-distribution but **degraded to 25% and 75% accuracy** on unseen LLMs, quantifying how severely lexical models overfit to GPT-specific artifacts.
- Extracted 11 stylistic features across 1,460 essays, establishing that AI text carries a higher complex-sentence ratio (0.84 vs. 0.53), greater vocabulary richness (0.59 vs. 0.42), and a **37-point lower Flesch Reading Ease** (20.9 vs. 58.0) than human writing.
- Diagnosed model confidence with reliability curves, confusion matrices, and gradient-based token attribution, documenting mid-range miscalibration and the false-positive risk that detection tools pose to non-native English writers.

Patch-Based Detection of AI-Generated Facial Images

Spring 2026

Python, PyTorch, CNNs, NumPy, Google Colab (A100)

- Trained and evaluated 5 CNN and feedforward architectures on **140,000 StyleGAN and Flickr face images**, holding compute constant across patch scales by tuning batch size so every model ran 6,950 steps per epoch under validation-loss early stopping.
- Demonstrated that majority-vote aggregation over 32×32 patches raises accuracy from **86.85% per patch to 99.17% per image**, beating the full-image 128×128 CNN baseline (97.10%) and confirming that patch-level errors are largely uncorrelated within an image.
- Quantified the accuracy–compute tradeoff driving model selection: the 2% aggregate gain cost **5.6× more training time** (3,367s vs. 596s), and under a 10% training-set restriction full-image CNNs retained more accuracy than patch models (90.37% vs. 86.02%).
- Built confidence-weighted patch heatmaps that localize GAN artifacts to hair boundaries and background transitions, adding spatial interpretability that binary whole-image classifiers cannot provide.

EXPERIENCE

Cybersecurity Intern

Jun. 2024 – Aug. 2024, Jun. 2025 – Aug. 2025

Aspire Technologies and Solutions

Manhattan, NY

- Monitored SIEM dashboards and triaged anomalous network activity alongside the security team, contributing to a 20% reduction in average incident response time over the internship period.
- Onboarded and trained 3+ incoming interns on internal tooling and security protocols, cutting ramp-up time by roughly 30% and shortening time-to-first-ticket.

Shift Manager

Feb. 2022 – Aug. 2025

Lea Restaurant

Brooklyn, NY

- Analyzed sales data and peak-hour workload patterns to restructure menu offerings and staff scheduling, increasing sales 10% in one quarter and reducing overtime costs 20% while managing a team of 5+.

PROJECTS

EcoHealth AI | *Python, Flask, MongoDB, OpenAI API*

November 2024

- Led a 4-person team to **3rd place out of a 24-hour hackathon field**, building a full-stack digital twin that ingests real-time user data across transportation, diet, and wellness to score environmental impact.
- Engineered Flask REST APIs backed by MongoDB and an LLM recommendation layer, delivering personalized interventions from live user metrics.

Class Registration Algorithm | *Java*

December 2023

- Built a constraint-satisfaction scheduler over **1,000+ student records** parsed from CSV, resolving 95% of course conflicts under Vassar's preregistration rules.
- Profiled and refactored the matching loop for a 20% runtime improvement, validated by a regression test suite that drove scheduling errors to zero.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, Transformers (DistilBERT), CNNs, scikit-learn, NLP, Stylometric & Feature Analysis, Model Evaluation (F1, ROC, Calibration), Prompt Engineering, LLM APIs

Data & Analysis: Pandas, NumPy, Matplotlib, TF-IDF / Bag-of-Words, CSV & Large-Scale Dataset Pipelines, Statistical Inference, RStudio

Languages: Python, Java, C, OCaml, Assembly, HTML/CSS, SQL

Tools: Git, Google Colab, VS Code, IntelliJ, Flask, MongoDB, Android Studio

Spoken Languages: English, Bengali, Spanish